

Nusrat Islam

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BIOGRAPHY

I am an M.Sc. student in Statistics and Data Science at Islamic University, Bangladesh, and a Research Assistant at the Statistical Intelligence Research Lab, with research experience in data-driven modeling, machine learning, deep learning, and AI for healthcare. My research has resulted in peer-reviewed publications in PLOS ONE and IEEE conferences, and I have been actively involved in research on medical image analysis, computer vision, and public health data analytics. My current work focuses on developing novel activation functions for deep learning.

RESEARCH INTERESTS

Data Science · Deep Learning · Medical Image Analysis · Computer Vision · Machine Learning · Optimization Algorithms · AI in Healthcare.

EDUCATION

M.Sc. – Islamic University, Bangladesh <i>Statistics and Data Science; Ongoing</i>	Kushtia-7003, Bangladesh <i>Jan 2026 – Present</i>
B.Sc. – Islamic University, Bangladesh <i>Statistics; CGPA: 3.81/4.00 (3rd among 50 peers)</i> <i>Last Semester: 4.00/4.00</i>	Kushtia-7003, Bangladesh <i>Jan 2020 – Nov 2025</i>
Govt. Pioneer Girls' College <i>H.S.C.; Science; GPA: 5.00/5.00</i>	Khulna-9100, Bangladesh <i>Jun 2017 – Jul 2019</i>
Uttar Khulna S.M.A. Majid Secondary School <i>S.S.C.; Science; GPA: 5.00/5.00</i>	Khulna-9231, Bangladesh <i>Jan 2012 – May 2017</i>

RELEVANT COURSEWORK

Data Science & Data Mining · R and Python for Data Science · Multivariate Analysis · Bioinformatics & Biostatistics · Regression Analysis · Bayesian and Classical Inference · Estimation Theory & Test of Hypothesis · Probability Theory · Calculus & Stochastic Process · Sampling Technique.

PUBLICATIONS

Journal Papers

1. Suman Biswas, Md. Mahamudul Islam, **Nusrat Islam**, Md. Abdur Rahim Mia, “National data meets AI: Machine learning for predicting overweight/obesity among ever-married Bangladeshi women,” *PLOS ONE*.
DOI: [10.1371/journal.pone.0341821](https://doi.org/10.1371/journal.pone.0341821)

Conference Papers

1. **Nusrat Islam**, Fahad Ibne Rahman, Suman Biswas, “SNM-TransNet: A Novel SNMish Activation-Based Attention-Enhanced Hybrid Architecture for Accurate Breast Cancer Image Classification,” *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence and Networking (QPAIN 2026)*, *IEEE Xplore*.
DOI: [10.1109/QPAIN69676.2026.11546422](https://doi.org/10.1109/QPAIN69676.2026.11546422)
2. Fahad Ibne Rahman, **Nusrat Islam**, Md. Khairul Islam, “FocalNet: An Attention-Free Novel Architecture for Brain Stroke Classification Using CT Imaging,” *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence and Networking (QPAIN 2026)*, *IEEE Xplore*.
DOI: [10.1109/QPAIN69676.2026.11545628](https://doi.org/10.1109/QPAIN69676.2026.11545628)
3. Fahad Ibne Rahman, **Nusrat Islam**, Md Emamul Hossen, Md. Khairul Islam, “A Deep Learning Approach Based on XAI and ViT-GRU Hybrid Model for Brain Tumor Classification Using MRI Images,” *IEEE 1st International Conference on Quantum Photonics, Artificial Intelligence and Networking (QPAIN 2025)*, *IEEE Xplore*.
DOI: [10.1109/QPAIN66474.2025.11171653](https://doi.org/10.1109/QPAIN66474.2025.11171653)

MANUSCRIPTS UNDER REVIEW

Journal Papers

1. Fahad Ibne Rahman, Nayan Kumar Sarkar, **Nusrat Islam**, Md. Ripaj Uddin, “Discovery of Wee1 Kinase Inhibitors from *Mangifera indica* Leaf Phytochemicals Ellagic Acid, Vitexin, and Isoswertisin for Colon Cancer: An Integrated Computational Approach,” *Phytomedicine (Elsevier)*, *CiteScore: 13.2*, *Impact Factor: 11.3*.

MANUSCRIPTS IN PREPARATION

1. **Nusrat Islam**, Fahad Ibne Rahman, “PVTv2-DVRNet: A Dynamic Vessel Refinement Network for Accurate Retinal Vessel Segmentation in Fundus Images,” *Manuscript in Preparation–Submitted for NST Fellowship 2026-27 Fiscal Year*.
2. Fahad Ibne Rahman, **Nusrat Islam**, “STanh: A Novel Self-Gated Activation Function for Robust and Efficient Deep Learning Across Convolutional and Transformer Architectures,” *Manuscript in Preparation–2026*.
3. Fahad Ibne Rahman, **Nusrat Islam**, “FPVT-UNet: A Multi-Scale Parallel Focal Modulation-PVTv2 Bottleneck Architecture for Automated Brain Stroke Lesion Segmentation from NCCT Images,” *Manuscript in Preparation–2026*.

RESEARCH EXPERIENCE

- **Statistical Intelligence Research Lab** Islamic University, Bangladesh
Research Assistant Sep 2023 – Present
 - Director: Suman Biswas, Assistant Professor, Department of Statistics and Data Science, Islamic University, Bangladesh.
 - Research Area: Data Analytics, Machine Learning, Deep Learning, Data Science, Public Health.
- **Bangladesh Academy for Rural Development** Kotbari, Comilla-3503, Bangladesh
Fieldwork 19–23 Feb 2025
 - Supervisor: Md. Kamal Uddin, Professor, Department of Statistics and Data Science, Islamic University, Bangladesh.
 - About: Bangladesh Academy for Rural Development (BARD) provided hands-on fieldwork experience in rural socio-economic research and primary data collection. Under BARD’s operational framework, I collected real-world data from rural communities in its surrounding areas, working collaboratively with fellow students and faculty members. The collected data were analyzed using statistical and machine learning techniques, culminating in a structured and comprehensive fieldwork defense.

UNDERGRADUATE PROJECT

SNM-TransNet: A Novel Dual SNMish Activation-Based Hybrid Deep Learning Framework for Histopathological Breast Cancer.

Supervisor: [Suman Biswas](#), Assistant Professor, Department of Statistics and Data Science, Islamic University, Bangladesh

HONORS AND AWARDS

1. **University Merit Scholarship** (B.Sc.) – 2022 & 2024
2. **University General Scholarship** (B.Sc.) – 2023
3. **National Board Scholarship** – Secondary School Certificate Examination, 2017.
4. **Best Presenter Award** – “SNM-TransNet: A Novel SNMish Activation-Based Attention-Enhanced Hybrid Architecture for Accurate Breast Cancer Image Classification,” *IEEE QPAIN 2026, CUET, Chattogram, Bangladesh. – 2026*
5. **Best Presenter Award** – “FocalNet: An Attention-Free Novel Architecture for Brain Stroke Classification Using CT Imaging,” *IEEE QPAIN 2026, CUET, Chattogram, Bangladesh. – 2026*

TECHNICAL SKILLS

Languages: Python, R

Frameworks/Libraries: TensorFlow, Keras, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Tools & Platforms: LaTeX, SPSS, SAS, GitHub, Kaggle, Microsoft Office (Word, PowerPoint, Excel)

Databases: MySQL

Soft Skills: Leadership, Project Management, Time Management, Writing, Public Speaking

LEADERSHIP AND VOLUNTEERING

- **Instructor – Python for Data Science: Beginners to Advance Course** Dec 2025 – Jun 2026
 - * This course was organized by the Department of Statistics and Data Science, Islamic University, Bangladesh. It was open to all students of Islamic University, Bangladesh. The main focus of the course was how Python programming can be applied to Data Science, covering practical implementation and hands-on applications.

REFEREES

Suman Biswas

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Prof. Dr. Md. Kamal Uddin

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